CECILIA LIN

New York, NY | xixilin.dev | LinkedIn | GitHub | U.S. Permanent Resident

CAREER SUMMARY

Data analyst with hands-on experience in statistical modeling, geospatial analysis, and data visualization. MS Business Analytics candidate at Baruch Zicklin with a concentration in Data Analytics. Skilled in Python, R, SQL, and building interactive analytical dashboards.

EDUCATION

Zicklin School of Business, Baruch College *Expected May 2027*
M.S. Business Analytics, Concentration in Data Analytics

Georgia Institute of Technology Admitted, starting Spring 2027
M.S. Computer Science (Online)

City College of New York, CUNY *May 2025*
Bachelor of Arts, Concentration in Digital Design

TECHNICAL SKILLS

Python | R | SQL | Excel | NumPy | Pandas | SciPy | ggplot2 | Plotly Dash | Tableau | Microsoft Access | Git/GitHub

PROJECTS

NYC Property Market Dashboard | Python · Plotly Dash · Pandas · SciPy · Render May 2026
Live Demo | Source Code

- Combined 65 yearly borough sales files (2013–2025, 1.1M rows) with a second sales dataset, handling header and borough-name changes across years
- Cleaned the data to 498K residential sales by removing invalid prices, non-residential, whole-building and hotel condo sales, and duplicates
- Joined sales with ZIP-level income, school rankings, NYPD arrest and CompStat crime data, and MTA subway stations; used a SciPy KD-Tree to assign 279K arrests to ZIP codes and compute each ZIP's subway distance
- Built the dashboard layout and linked filters across 7 charts; reduced the deployed data file to about 2 MB and deployed on Render

Walmart Retail Sales Performance Analysis | R · Linear Regression · Lasso/Ridge · ARIMA May 2026
Source Code

- Analyzed 6,434 weekly sales records across 45 Walmart stores using association analysis and multiple linear regression, building and comparing 3 models (baseline, full, reduced) to identify drivers of retail performance
- Found macroeconomic variables explained only about 2.5% of sales variance, while adding store as a factor raised R^2 to about 92%
- Tested Lasso and Ridge regression and built an ARIMA model for a 12-week sales forecast

BrightSmile Dental | Microsoft Access · SQL · ERD Dec 2025
Video Demo | Source Code

- Led a team building a relational database for a dental clinic case study, translating requirements from 4 staff roles into a schema normalized to 3NF
- Co-designed the ER model; built the tables, login and data-entry forms, queries for revenue by procedure and low-stock supplies, and monthly and provider reports

PROFESSIONAL EXPERIENCE

Sinovision, Inc. | Queens, NY Aug 2017 – Aug 2024
Broadcasting Technician

- Built and ran daily broadcast playlists in Crispin automation software, scheduling programs, ads, and station IDs across multiple channels
- Operated live event broadcasts on LiveStream equipment, including annual Chinese New Year programs, with no room for timing errors
- Monitored on-air signals and troubleshot playout issues, escalating to the IT team and WNBC when needed